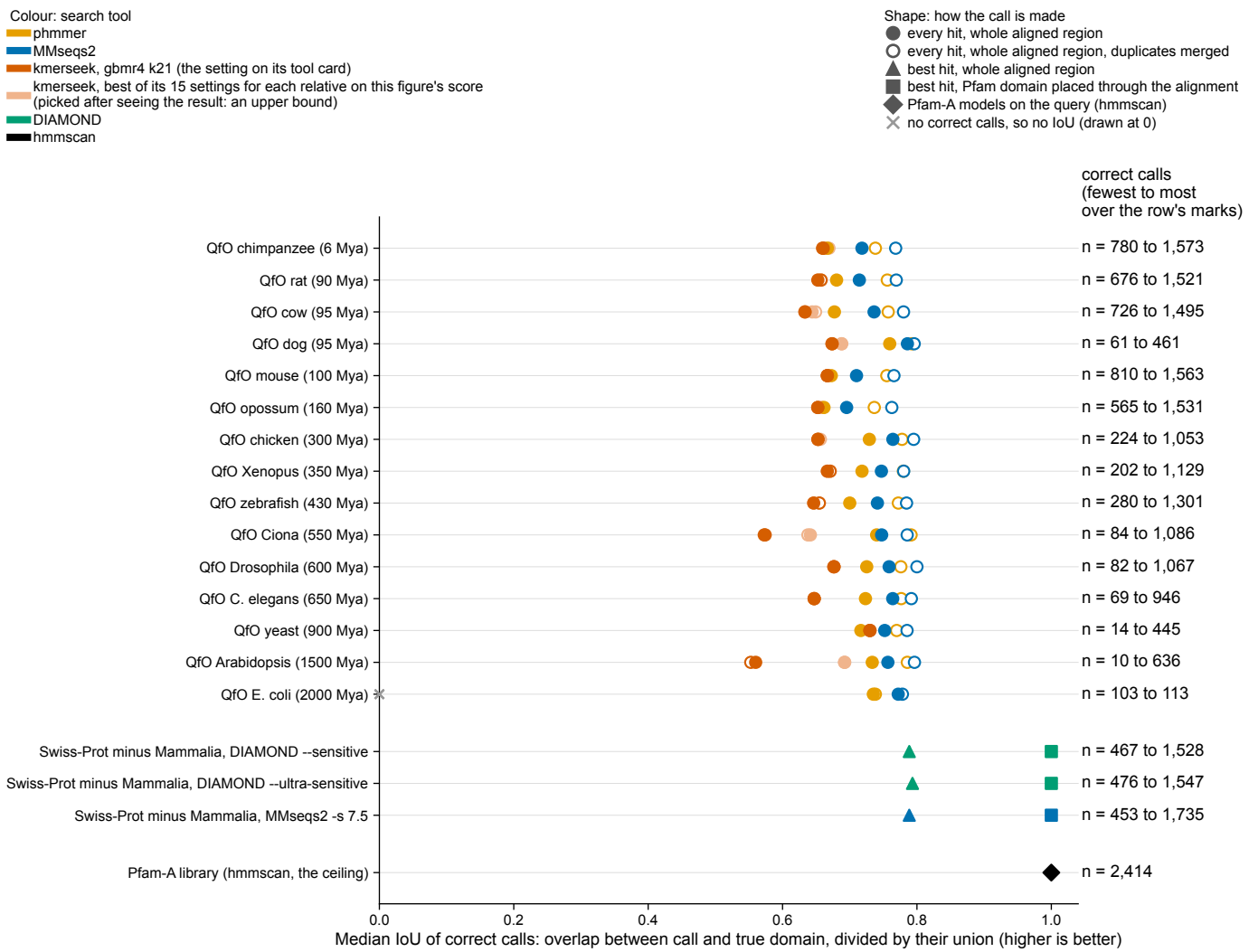
Colour: search tool
phmmer
MMseqs2
kmerseek, gbmr4 k21 (the setting on its tool card)
kmerseek, best of its 15 settings for each relative on this figure's score (picked after seeing the result: an upper bound)
DIAMOND
hmmscan
Shape: how the call is made
every hit, whole aligned region
every hit, whole aligned region, duplicates merged
best hit, whole aligned region
best hit, Pfam domain placed through the alignment
Pfam-A models on the query (hmmscan)
no correct calls, so no IoU (drawn at 0)
correct calls (fewest to most over the row's marks)
QfO chimpanzee (6 Mya)
QfO rat (90 Mya)
QfO cow (95 Mya)
QfO dog (95 Mya)
QfO mouse (100 Mya)
QfO opossum (160 Mya)
QfO chicken (300 Mya)
QfO Xenopus (350 Mya)
QfO zebrafish (430 Mya)
QfO Ciona (550 Mya)
QfO Drosophila (600 Mya)
QfO C. elegans (650 Mya)
QfO yeast (900 Mya)
QfO Arabidopsis (1500 Mya)
QfO E. coli (2000 Mya)
Swiss-Prot minus Mammalia, DIAMOND --sensitive
Swiss-Prot minus Mammalia, DIAMOND --ultra-sensitive
Swiss-Prot minus Mammalia, MMseqs2 -s 7.5
Pfam-A library (hmmscan, the ceiling)
n = 780 to 1,573
n = 676 to 1,521
n = 726 to 1,495
n = 61 to 461
n = 810 to 1,563
n = 565 to 1,531
n = 224 to 1,053
n = 202 to 1,129
n = 280 to 1,301
n = 84 to 1,086
n = 82 to 1,067
n = 69 to 946
n = 14 to 445
n = 10 to 636
n = 103 to 113
n = 467 to 1,528
n = 476 to 1,547
n = 453 to 1,735
n = 2,414
0.0
0.2
0.4
0.6
0.8
1.0
Median IoU of correct calls: overlap between call and true domain, divided by their union (higher is better)